

All instances of a class have the same instance variables, with different values inside them. The state or current values for an instance of a class is called the “state” of that class. The current values of those variables define the current situation or state of this instance of the class. For example, a class called `HourlyEmployee` contains instance variables:

```
first_name
last_name
soc_sec_number
hourly_rate
current_vacation_time
```

All objects that are instances of the same class share the same behaviour. They all have the same methods. We could send the same messages to all instances of the a class and all would understand and respond to the messages.

Our class is: `HourlyEmployee`

All instances of this class have these methods:

```
calculate_pay()
setName()
getName()
setSSN()
getSSN()
getVacationTime()
setVacationTime()
getHourlyRate()
setHourlyRate()
```

Every example, or instantiation, of this class has the same methods (behaviour) available to it.

Let's instantiate `HourlyEmployee` :

```
HourlyEmployee joseph; // empty reference.
joseph = new HourlyEmployee('Joe' 'Smith', '598-22-7893', '$10.00', '22.25');
```

Now, we've created an instance of the class `HourlyEmployee`. Our class is: `HourlyEmployee`. We have instantiated `HourlyEmployee` and our instance is called `joseph`. The identity of the instance is `joseph`.

The state of the instance is: